

LP- II_2025-26_Problem statements (List) for Mock-II and PR Exam

Sr. No	Problem Statement
1	Implement depth first search algorithm and Breadth First Search algorithm, Use an undirected graph and develop a recursive algorithm for searching all the vertices of a graph or tree data structure.
2	Implement A star Algorithm for any game search problem.
3	Implement a solution for a Constraint Satisfaction Problem using Branch and Bound and Backtracking for n-queens problem or a graph coloring problem
4	Develop an elementary Chatbot for any suitable customer interaction application (Rule based & GUI Based).
5	Implement Greedy search algorithm for any three of the following application: I Selection Sort II Minimum Spanning Tree III Single-Source Shortest Path Problem IV Job Scheduling Problem V Prim's Minimal Spanning Tree Algorithm VI Kruskal's Minimal Spanning Tree Algorithm VII Dijkstra's Minimal Spanning Tree Algorithm
6	Implement any one of the following Expert System I Information management II Hospitals and medical facilities III Help desks management IV Employee performance evaluation V Stock market trading VI Airline scheduling and cargo schedules
7	CC- Design and deploy a static website on a cloud virtual machine/Instance. Configure the server and networking settings so that the website is publicly accessible and can be updated remotely. OR IS- Write a Java/C/C++ program to read a plaintext string and the mod factor from the user, encrypt it using a bitwise logical operator and display the ciphertext.

8	CC- Secure File Sharing Between Cloud Instances Create a cloud-based environment where two virtual machines communicate securely within a virtual network. Implement a mechanism to transfer files between the instances while ensuring proper access permissions. OR IS- Write a Java/C/C++ program to perform encryption and decryption using columnar Transposition and rail fence method.
9	CC- Online Blog Application Deployment Deploy a full-stack blog application (frontend, backend, and database) on a cloud platform. Configure the server environment so that users can create, view, and manage blog posts through a web interface. (Note:- Take GitHub Repository / dummy application) OR IS- Write a Java/C/C++/Python program to implement a simplified DES algorithm and demonstrate encryption and decryption process.
10	CC- Cloud-Based Student Record Management System Deploy a web application that stores and manages student records using a database and backend API. Deploy the frontend interface and ensure users can perform operations such as adding, updating, and retrieving student data. (Note:- Take GitHub Repository / dummy application) OR IS- Write a Java/C/C++/Python program to implement the RSA algorithm for encryption and decryption of text file.
11	CC- Deployment of an E-Commerce Web Application (Full Stack Mern Application) Deploy a sample e-commerce application on a cloud virtual machine. Configure the backend services and database so that users can browse products and simulate basic purchase functionality. (Note:- Take GitHub Repository / dummy application) OR IS - Demonstrate puzzle friendly property of Hashing algorithm MD5 using Proof of Work technique in Bitcoin mining in JAVA.

12	CC- Static Website with Domain and Public Access Deploy a static website on a cloud server and configure the network settings so that it is accessible through the server's public IP or domain name. OR IS- Write a Java/C/C++ program to read a plaintext string and the mod factor from the user, encrypt it using a bitwise logical operator and display the ciphertext.
13	CC- MERN-Based Task Management Application Deploy a MERN-based task management system where users can create, update, and delete tasks. Ensure the frontend communicates with the backend API and the database is hosted properly. OR IS- Write a Java/C/C++ program to perform encryption and decryption using columnar Transposition and rail fence method.
14	CC- Cloud-Based To-Do List Manager (Full Stack Mern Application) Deploy a task management application that allows users to create, update, and delete tasks. Configure the application so that it stores data in a backend database hosted on the cloud. OR IS- Write a Java/C/C++/Python program to implement a simplified DES algorithm and demonstrate encryption and decryption process.
15	CC- Online Event Registration System (Full Stack Mern Application) Deploy a web application where users can register for an event by submitting their details, which are stored in a cloud-hosted database. (Note:- Take GitHub Repository / dummy application) OR IS- Write a Java/C/C++/Python program to implement the RSA algorithm for encryption and decryption of text file.
16	CC- Student Record Management System Develop an application in Salesforce using the Apex programming language and visual force page to manage student records (Name, Roll No, Class, Mobile No) stored in a custom object. OR IS - Demonstrate puzzle friendly property of Hashing algorithm MD5 using Proof of Work technique in Bitcoin mining in JAVA.

17	CC- Employee Management System Develop a menu-driven Employee Management System that manages employee records (Emp ID, Emp Name, email, birth date, Department) in Salesforce using console based. OR IS- Write a Java/C/C++ program to read a plaintext string and the mod factor from the user, encrypt it using a bitwise logical operator and display the ciphertext.
18	CC- Bank Account System Develop a menu-driven Bank Account System that manages Customer records (Emp ID, Emp Name, email, birth date, Department) in Salesforce using console based. OR IS- Write a Java/C/C++ program to perform encryption and decryption using columnar Transposition and rail fence method.
19	CC- Product Inventory Management System Develop an application in Salesforce using the Apex programming language and visual force page to manage Product Inventory records (Product Name, Serial No, Manufacture Date, Expiry Date) stored in a custom object. OR IS- Write a Java/C/C++/Python program to implement a simplified DES algorithm and demonstrate encryption and decryption process.
20	CC- Develop an Employee Management Lightning Web Component that allows users to add employee records with validation. The system should validate the following before saving: ● Employee Name cannot be empty and must contain at least 3 characters. ● Employee ID must be greater than 0 and should be unique. ● Salary must be greater than 10,000 and less than 500,000. ● Email addresses must follow a valid email format. ● The department must be selected from the available list. ● Joining Date cannot be a future date. OR IS - Demonstrate puzzle friendly property of Hashing algorithm MD5 using Proof of Work technique in Bitcoin mining in JAVA.

21	CC- Develop an console based Apex program that sends an email notification to a specified email address using Salesforce email services. The program should define the recipient email, subject, and message body, and send the email using the built-in messaging class. (with /without Attachment) OR IS- Write a Java/C/C++ program to read a plaintext string and the mod factor from the user, encrypt it using a bitwise logical operator and display the ciphertext.												
22	CC- Develop an Apex program that sends an email notification to a specified email address using Salesforce email services. The program should define the recipient email, subject, and message body, and send the email (with /without Attachment) using the built-in messaging class. Give the appropriate message on invalid email id (with frontend using visualforce pages) OR IS- Write a Java/C/C++ program to perform encryption and decryption using columnar Transposition and rail fence method.												
23	CC- Develop a College Management Lightning Application to store and manage student and faculty records. The system should ensure data accuracy using validation rules while creating or updating records. The system should validate the following before saving: <table border="1" data-bbox="228 1075 1500 1524"> <thead> <tr> <th data-bbox="228 1075 810 1115">Validation for students-</th><th data-bbox="810 1075 1500 1115">Validation for Faculty-</th></tr> </thead> <tbody> <tr> <td data-bbox="228 1115 810 1188">Marks should be between 0 and 100.</td><td data-bbox="810 1115 1500 1188">Faculty Name cannot be empty and must contain at least 3 characters.</td></tr> <tr> <td data-bbox="228 1188 810 1262">Roll number must be greater than 0</td><td data-bbox="810 1188 1500 1262">Faculty ID must be greater than 0 and should be unique.</td></tr> <tr> <td data-bbox="228 1262 810 1335">Ensure the email contains @ symbol.</td><td data-bbox="810 1262 1500 1335">Salary must be greater than 10,000 and less than 500,000.</td></tr> <tr> <td data-bbox="228 1335 810 1451">The student's name cannot be blank.</td><td data-bbox="810 1335 1500 1451">The department must be selected from the available list.</td></tr> <tr> <td data-bbox="228 1451 810 1524"></td><td data-bbox="810 1451 1500 1524">Joining Date cannot be a future date.</td></tr> </tbody> </table> OR IS- Write a Java/C/C++/Python program to implement the RSA algorithm for encryption and decryption of text file.	Validation for students-	Validation for Faculty-	Marks should be between 0 and 100.	Faculty Name cannot be empty and must contain at least 3 characters.	Roll number must be greater than 0	Faculty ID must be greater than 0 and should be unique.	Ensure the email contains @ symbol.	Salary must be greater than 10,000 and less than 500,000.	The student's name cannot be blank.	The department must be selected from the available list.		Joining Date cannot be a future date.
Validation for students-	Validation for Faculty-												
Marks should be between 0 and 100.	Faculty Name cannot be empty and must contain at least 3 characters.												
Roll number must be greater than 0	Faculty ID must be greater than 0 and should be unique.												
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